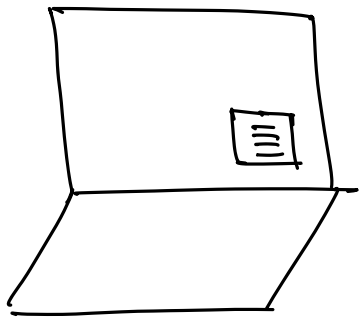
Ollama.

↳ Ollama is a tool that allows us to download, run & interact with large language models locally on our machine.

Model
- Open Source
- Proprietary.

Meta llama,
Deepseek, ...



Model : 16 Billion parameters.

Billion → 10^9

parameters → 4 Bytes parameter size.

$16 \times 10^9 \times 4$ Bytes

64×10^9 Bytes

64 GB.

10^3 Bytes = 1KB

10^3 KB = 1MB

10^3 MB = 1GB

Advantages of running LLM locally :-

→ Data security & privacy.

→ Offline usage.

→ Faster than external LLM.

→ Better for fine-tuning

Ollama is the easiest way to get up and running with large language models such as gpt-oss, Gemma 3, DeepSeek-R1, Qwen3 and more.

ollama run gemma3:1b

Model Quantization

Quantization means representing model parameters & weight in lower precision data types so that the model uses lesser memory & runs faster.

⇒ LLM are huge, and full precision model may be too large for many local machines so using Quantization, the size of LLM can be reduced & we can run that LLM in local m/c.

↳ Less memory usage.

FP32 FP16 INT8 INT4 ...

Quantization is trade-off.

CPU vs GPU. Inference.

Generating output from the LLM.

⇒ RAM → Used by CPUs.

VRAM → GPUs.

Optimization Techniques.

→ Quantization

→ Reduce tokens & Context window.

→ Run model on GPU.